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PYTHON CURRICULUM

PYTHON INSTALLATION AND REFRESHERS

- Introduction to section
- Python Installation Overview
- Python Setup (Windows) / Python Setup (Mac)
- Python Integrated Development Environment Overview (VS Code)
- Setup Integrated Development Environment (VS Code)
- Download Python Refresher Source Code
- Variables in Python
- Comments

Assignment 1: _____

- Getting User Input
- String formatting in Python
- String Assignment
- String Assignment Solution (Together)
- Lists in Python
- Sets and Tuples
- Lists Assignment
- Lists Assignment Solution
- Booleans and Operators
- If Else Statements
- If Else Assignment
- If Else Assignment Solution
- Loops in Python

Assignment 2: Loops Assignment_____

- Dictionaries in Python
- Dictionaries Assignment
- Dictionaries Assignment Solution
- Functions in Python

Assignment 3: Functions Assignment_____

- Imports in Python
- Object Oriented Programming in Python
- Class Inheritance

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GETTING STARTED WITH FASTAPI

- What is FastAPI?
- Installing Python?
- Installing FastAPI and virtual environment installation (Windows)
- Installing FastAPI and virtual environment installation (Mac)
- Your first 'hello world' program using default GET.
- GET and PUT
- What is JSON format
- JSON response
- Practical GET and PUT
- Interactive API docs
- Alternative API docs
- What is uvicorn
- Features of using FastAPI

PYTHON HINT TYPES

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- Python Type hint's introduction
- Why they are important
- Using hint types in function
- Specifying function return type

FASTAPI

- Import FastAPI
- Creating FastAPI Instance
- URL
- HTTP methods
- What is decorator
- Defining a decorator
- Defining path operation function
- Path parameter
- Path parameter with types
- Working with python enumeration
- Query parameters
- Multiple path and query parameters
- Request Body
- Query Parameters and String Validations
- Form data
- Request files
- Practical Demo Request forms and files
- Handling errors

GIT- VERSION CONTROL

- Git Introduction
- Git Windows Installation
- Git Mac Installation
- Git Basics Overview
- Git Basics Practice
- Git Branches Overview
- Git Branches Practice
- GitHub Intro
- Git Remote Commands Slides
- GitLab: Create Account & Remote Repository
- GitLab: Use SSH key

PROJECT SETUP

- Introduction to python virtual environments
- Introduction to poetry
- Install poetry on Windows / Mac
- Initialize project with poetry
- Initialize project using git and Gitlab

FASTAPI REQUEST METHOD LOGIC

- GET Request Method Overview
- Create FastAPI Endpoint
- FastAPI Project: Enhance Get Request
- FastAPI Project: Path Parameters Overview
- FastAPI Project: Path Parameters
- FastAPI Project: Query Parameters Overview
- FastAPI Project: Query Parameters
- FastAPI Project: Post Request Overview
- FastAPI Project: Post Request
- FastAPI Project: Put Request Overview
- FastAPI Project: Put Request
- FastAPI Project: Delete Request Overview
- FastAPI Project: Delete Request

Assignment: FastAPI Assignment

MOVE FAST WITH FASTAPI

- FastAPI Project: Setup Books 2 Project
- FastAPI Project: Post Request before Validation
- FastAPI Project: Pydantics and Data Validation Overview

- FastAPI Project: Pydantic Book Request Validation
- FastAPI Project: Fields - Data Validation
- FastAPI Project: Pydantic Configurations
- FastAPI Project: Fetch Book
- FastAPI Project: Fetch Books by Rating
- FastAPI Project: Update Book with Put Request
- FastAPI Project: Delete Book with Delete Request
- FastAPI Project: Data Validation Path Parameters
- FastAPI Project: Data Validation Query Parameters
- FastAPI Project: Status Codes Overview
- FastAPI Project: HTTP Exceptions
- FastAPI Project: Explicit Status Code Responses

SETUP DATABASE

- SQL Database Introduction with PostgreSQL
- Database Connection with ORM SQLAlchemy
- Database Tables (Models)
- Main (Create Database Connection for API)
- SQL Queries Introduction

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API REQUEST METHODS

- FastAPI Project: Get All Todos from Database
- FastAPI Project: Get Todo by ID
- FastAPI Project: Post Request (Todo Project)
- FastAPI Project: Put Request (Todo Project)
- FastAPI Project: Delete Request (Todo Project)

AUTHENTICATION & AUTHORIZATION

- FastAPI Project: Starting Authentication & Authorization
- FastAPI Project: Routers Scale Authentication File
- FastAPI Project: Router Scale Todos File
- FastAPI Project: One to Many Relationship
- FastAPI Project: Foreign Keys
- FastAPI Project: Users Table Creation
- FastAPI Project: Create First User
- FastAPI Project: Hash User's Password
- FastAPI Project: Save User to Database
- FastAPI Project: Authenticate a User
- JSON Web Token (JWT) Overview
- FastAPI Project: Encode a JSON Web Token (JWT)
- FastAPI Project: Decode a JSON Web Token (JWT)
- FastAPI Project: Authentication Enhancements

AUTHENTICATE REQUESTS

- FastAPI Project: Post Todo (User ID)
- FastAPI Project: Get All Todos (User ID)
- FastAPI Project: Get Todo (ID + User ID)
- FastAPI Project: Put Todo (User ID)
- FastAPI Project: Delete Todo (User ID)
- FastAPI Project: Admin Router

WORKING WITH FILES

- File
- UploadFile
- Checking file type
- Making files statically available
- Downloading files

DATABASE WITH SQLALCHEMY

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- Quick intro about Dependencies
- Databases in FastAPI
- Boilerplate code
- Create database and table
- Solving a common python environment problem
- Create database and table continued
- Write data in database
- Process review
- Create and read
- Update and delete
- Relationships

LARGE PRODUCTION DATABASE SETUP

- Production DBMS
- PostgreSQL Introduction
- PostgreSQL Windows Installation
- PostgreSQL Mac Installation
- PostgreSQL Create Database Table
- PostgreSQL Connect to FastAPI

ALEMBIC DATA MIGRATION

- Alembic Data Migration Overview
- Alembic Introduction
- Quick starting notes

- Alembic Setup
- Alembic Upgrade Command
- Alembic Add New Column
- Alembic Create New Database Table
- Alembic Create FK Relationship
- Alembic Enhance Application with New API

DEPLOYING FASTAPI APPLICATIONS

- Deployment: Render Introduction
- Deployment: Introduction to docker and docker-compose
- Deployment: Add Requirements File
- Deployment: Render Setup
- Deployment PostgreSQL: Production Database
- Deployment: Deploy application using docker

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EVALUATION

Task 1: Sending Email using FastAPI Framework in Python

Task 2: How to Generate JWT Tokens using FastAPI in Python

Task 3: Build a CRUD App with FastAPI and SQLAlchemy

Task 4: Implement a REST API service by creating a blog service in FastAPI